

### Class Plan

**Section #:** 44109

**Semester:** Fall (Sept-Dec) 2026

**Professor:** Jon Berge, Ph.D. P.Eng.

**Program(s):** Bachelor of Engineering (Mechanical), Bachelor of Engineering (Electrical)

### **Evaluation Plan**

Students demonstrate their learning in the following ways:

|                                                     |      |
|-----------------------------------------------------|------|
| In-class Activity Submissions (8 @ 2.5% each)       | 20%  |
| Individual Essay Assignment Proposal                | 2.5% |
| Individual Essay Assignment Progress Report         | 2.5% |
| Individual Essay Assignment Peer Teach Presentation | 15%  |
| Individual Essay Assignment Peer Teach Handout      | 10%  |
| Individual Essay Assignment Essay                   | 30%  |
| Final Test                                          | 20%  |

#### In-class Activities:

Submission of in-class activities is done during scheduled lectures and is based on class and small group discussion of the topics covered in class. Students must be present in class in order to submit work. All lectures will have graded in-class activities. The in-class activity component of the final grade will be calculated based on each individual student's best 8 submissions. Participation in the in-class activity involves both participation in the class discussion as well as submission of hand-written work. Students are to arrive to each class prepared with a working pen and at least 1 8½x11 sheet of paper (pages torn from notebooks are not acceptable).

#### Individual Essay Assignment:

Students will choose a contemporary example that illustrates one or more of the course themes. Based on this example students will complete the 3 parts of the project (essay, presentation and handout), more details will be provided in a separate document.

#### Final Test:

The comprehensive Final Test will assess students' learning of the subject matter presented in class.

## Weekly Timetable

| Date                          | Topics                                                                                                                        |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Week #1 (September 10)</b> | World-views, Preconceptions, Biases and How to Change your Mind<br>The 'Life Cycle' Perspective and The Role of Public Policy |
| <b>Week #2 (September 17)</b> | Pollution and Climate Change                                                                                                  |
| <b>Week #3 (September 24)</b> | Nuclear Energy                                                                                                                |
| <b>Week #4 (October 1)</b>    | Infrastructure and Environmental Assessment                                                                                   |
| <b>Week #5 (October 8)</b>    | Social Impacts of Infrastructure                                                                                              |
| <b>Week #6 (October 15)</b>   | Biomass, Fossil Fuels                                                                                                         |
| <b>Week #7 (October 22)</b>   | Fossil Fuels                                                                                                                  |
| <b>October 26-30</b>          | <b><i>Reading Week – No Classes</i></b>                                                                                       |
| <b>Week #8 (November 5)</b>   | Food and Agriculture                                                                                                          |
| <b>Week #9 (November 12)</b>  | Technology and Healthcare                                                                                                     |
| <b>Week #10 (November 19)</b> | Information Technology, Information Security, Social Media and Surveillance                                                   |
| <b>Week #11 (November 26)</b> | Failures of Technology, Whistleblowing, Disclosure, and Unintended Consequences                                               |
| <b>Week #12 (December 3)</b>  | Individual Essay Assignment Peer Teach                                                                                        |
| <b>Week #13 (December 10)</b> | Individual Essay Assignment Peer Teach                                                                                        |
| <b>Week #14</b>               | <b>Final Test</b>                                                                                                             |

# Key Academic Policies

## Classroom Use of Technology:

This course is intended to teach students critical analysis skills, this is done in part through class discussion and engagement. Typing notes in class has been found to reduce cognitive engagement with material [1-3] this impedes the effective teaching and learning of critical analysis. For this reason, the use of laptops and other digital devices with keyboards is not permitted in class. If students wish to take notes during lectures they are encouraged to handwrite limited notes using a tablet and stylus or pen and paper. All lecture slides are provided ahead of the delivery of lectures. The scope of the final test is limited to material found directly in the lecture slides and assigned readings. *If you have approved accommodations that you believe may be impacted by this policy please speak with me privately.*

## Use of Generative AI:

All work submitted for evaluation in this course must be the original work of the student submitting it. The use of generative AI tools (Chat GPT, Claude etc.) to write or edit work is not permitted and constitutes a breach of academic integrity.

## MEET Missed Evaluations Process:

Please refer to the supporting documents posted in the Student Virtual Community on SLATE. Missed evaluations not approved under this process will be awarded a grade of zero.

## MEET Late Assignment Policy:

All assigned work due as advised by the instructor. Late submission will attract a 10% deduction per day, to a maximum of 30%. Assignments submitted after 3 days late will be awarded a grade of zero.

## Academic Integrity Policy:

Students are reminded about the importance of completing original work with academic integrity and of Sheridan's Academic Integrity Policy. Software to detect plagiarism as well as the use of generative AI may be used on any student work. It is expected that all work submitted in this course be entirely original work produced by the student submitting it. Failure to follow these guidelines will be addressed through Sheridan's Academic Integrity policy: <https://www.sheridancollege.ca/-/media/project/sheridan/shared/files/about/administration-and-governance/policies-and-accountability/policies-procedures/academic-integrity-policy>

Any student who has concerns with any of the above course policies or their implementation is encouraged to reach out to me privately.

[1] G. Marano, G. D. Kotzalidis, F. M. Lisci, M. B. Anesini, S. Rossi, S. Barbonetti, et al., "The Neuroscience Behind Writing: Handwriting vs. Typing-Who Wins the Battle?" *Life* (Basel), vol. 15, no. 3, p. 345, Feb. 22 2025.

[2] C. Hu, "Why writing by hand is better for memory and learning," *Scientific American*, Jul. 30, 2024. [Online]. Available: <https://www.scientificamerican.com/article/why-writing-by-hand-is-better-for-memory-and-learning/>

[3] N. Humlicek, "How teaching handwriting supports brain Development (Opinion)," *Education Week*, Aug. 18, 2026. [Online]. Available: <https://www.edweek.org/teaching-learning/opinion-how-teaching-handwriting-supports-brain-development/2026/08>